Surface mass balance inference based on a transient glacier evolution model inversion

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Abstract. The distributed surface mass balance (SMB) of mountain glaciers is the key forcing for glacier evolution and for projections of meltwater availability. Yet direct observations remain confined to sparse stake networks, and geodetic estimates rest on the assumption of a stationary ice flux. Here, we present a new SMB inference method, implemented within the open-source Instructed Glacier Model (IGM), that recasts the problem as a fully differentiable transient inversion. The scheme uses observed surface velocities to initialise a dynamically consistent glacier model, then inverts multi-temporal surface elevation changes to estimate the spatial distribution of mass balance that best explains the observed glacier evolution. We apply the method to three well-studied glaciers in the Alps—Argentière, Great Aletsch and Rhône—evaluating all three with a single, glacier-independent pipeline. Validating the inferred SMB against in situ stake measurements shows that the surface-velocity misfit tracks the SMB misfit. These results validate our method and render it transferable to any glacier with accurate multi-temporal DEMs and surface velocities, even in the absence of stake measurements, opening a path towards regional-scale, distributed SMB inference.

1 Introduction

Mountain glaciers are sensitive recorders of climate and a critical component of regional water resources, yet projecting their evolution requires their distributed surface mass balance (SMB)—the spatial pattern of accumulation and ablation—which remains poorly constrained at the global scale, particularly in remote parts of the world. This is in part because it relies on direct, in situ measurements using stake networks and snow pits that are labour-intensive, and hazardous in steep accumulation areas. As a result, it is currently deployed on less than 0.2 % of the world's glacierised area (Cogley et al., 2011). As an alternative, regional and global climate models provide continuous gridded SMB, but they require field calibration (Radić et al., 2014; Marzeion et al., 2012). Most glaciers therefore remain effectively unmonitored, and regional-to-global mass-balance assessments depend on extrapolation from a few benchmark sites or on model reconstructions (Hugonnet et al., 2021; Zemp et al., 2019). As remote-sensing products of glacier surface velocity (Millan et al., 2022), ice thickness (Farinotti et al., 2019; Millan et al., 2022) and elevation change (Hugonnet et al., 2021; Beraud et al., 2022) reach near-global coverage, the question is no longer whether the data exist, but how to combine them to infer physically consistent SMB fields.

1.1 Existing approaches

SMB can be derived directly from climate models that simulate surface energy balance and precipitation, but their coarse resolution and the simplified treatment of high-mountain microclimate bias precipitation phase, accumulation gradients and melt, so they rarely reproduce observed glacier change without additional bias correction (Marzeion et al., 2012; Rounce et al., 2020). Alternatively, sparse direct measurements are extended to distributed fields by homogenising them against decadal geodetic volume change, as in the long-running Glacier Monitoring Switzerland programme (GLAMOS; GLAMOS, 2021); although these yield a valuable observational reference, they remain affected by a large number of statistical assumptions.

The route most closely related to the present work exploits the mass-conservation equation, which relates the rate of surface-elevation change to the divergence of the ice flux and the SMB (Cuffey and Paterson, 2010). Given ice thickness and surface velocity, the vertically-integrated flux can be computed, differentiated spatially, and combined with an observed elevation-change field to provide an SMB estimate. From early flux-gate estimates of emergence velocity in the ablation zone (Berthier and Vincent, 2012; Brun et al., 2018), the approach has been generalised to fully distributed SMB over entire glaciers (Bisset et al., 2020; Miles et al., 2021; Pelto and Menounos, 2021; Van Tricht et al., 2021; Kneib et al., 2022). Its state-of-the-art realisation is that of Kneib et al. (2024), who reconstructed a high-resolution distributed SMB for Argentière Glacier (French Alps) over 2012–2021 from Pléiades elevation change and velocity, ground-penetrating-radar thickness and three ice-thickness models.

Two difficulties beset this route. First, the flux divergence is the spatial derivative of the product of two independently-measured noisy fields—surface velocities and glacier thickness—so it is dominated by small-scale noise, which can be suppressed by *post hoc* spatial filtering (Van Tricht et al., 2021; Bisset et al., 2020); yet smoothing a quantity that respects mass conservation locally redistributes mass between grid cells and violates the very balance the method enforces. Secondly, the flux divergence is computed at a single observational snapshot and assumed representative of the whole observation period, whereas glacier geometry, velocity and flux all evolve under ongoing climate forcing (Jóhannesson et al., 1989; Zekollari and Huybrechts, 2015); treating the flux as stationary introduces a bias that grows with the length of the window—a limitation Kneib et al. (2024) acknowledge explicitly. Per-glacier tuning of smoothing length scales, density assumptions and weighting schemes further hampers transfer to the regional scales most relevant for water-resource applications.

1.2 This study

Two features distinguish our formulation from flux-divergence inversions. First, the SMB is recovered by solving the forward mass-conservation equation rather than by differentiating observed fields: the flux divergence is recomputed at every time step from the evolving glacier state, so the inferred SMB is dynamically consistent over the whole window—rather than assuming a stationary flux—and mass is conserved at every step, with no *post hoc* smoothing of a noisy divergence field. Second, we parametrise the SMB as a one-dimensional function of surface elevation, evaluated at every grid cell to yield a 2D field; this reflects the first-order elevation control on alpine SMB and reduces the inversion to roughly ten unknowns, rendering the otherwise severely ill-posed problem well-constrained without site-specific smoothing length scales or density assumptions

(Sect. 2.4). The price—horizontal variability within an elevation band, such as the avalanche-fed hotspots resolved by Kneib et al. (2024), is not recovered—is a deliberate trade-off of sub-elevation detail for a transferable, tuning-free inversion, to which we return in Section 4.

The accuracy of our geometry-based inversion is ultimately controlled by that of the input DEMs. Near-global elevation-change products (Hugonnet et al., 2021) make the method broadly applicable, but their inaccuracies limit the fidelity of the recovered SMB. Dedicated multi-temporal DEMs deliver the highest accuracy, and the rapidly expanding archive of such data is what makes a regional-scale application possible. We validate the method with two-epoch DEMs on three well-monitored Alpine glaciers of contrasting size—Argentière (French Alps), Great Aletsch and Rhône (Swiss Alps)—all inferred by the *same* two-step pipeline, without per-glacier retuning: the basal-motion parameter is reduced to a single scalar fixed by an ensemble search on the surface-velocity misfit, and no thickness survey is assumed.

Section 2 details the two-step pipeline—the input and validation data (Sect. 2.2), data assimilation of the initial dynamical state, then SMB inference by automatic differentiation through the transient forward run; Section 3 presents the inferred SMB profiles and their validation against in situ stake measurements; and Section 4 discusses the implications for regional-scale SMB inversion.

2 Methods

2.1 Overview

The observational inputs to our method are two surface digital elevation models (DEMs) of the glacier, acquired at epochs t_1 and t_2 several years apart, together with a surface-velocity field observed near t_1. From these we infer the glacier's surface mass balance (smb) with a two-step pipeline (Fig. 1), implemented within the open-source Instructed Glacier Model (IGM; Jouvet et al., 2026). The elevation change between the two DEMs is the signal used to recover the smb (Sect. 2.2). At its core is the depth-integrated mass-conservation equation, which links this geometry change to the smb,

$$\frac{\partial H}{\partial t} = smb - \nabla \cdot \mathbf{q}, \qquad \mathbf{q} = H\,\bar{\mathbf{u}}, \tag{1}$$

where H is the ice thickness, smb is the surface mass balance, q is the flux and $\bar{\mathbf{u}}$ is the depth-averaged horizontal velocity. The velocity is supplied by IGM's deep-learning ice-flow emulator (Jouvet and Cordonnier, 2023; Rosier and Jouvet, 2026), trained on a large ensemble of higher-order flow solutions to return $\bar{\mathbf{u}}$ from the ice geometry and the flow parameters. IGM can evaluate the emulator in two modes—online, in which its weights are updated during the simulation, and offline, in which they are frozen; we use the offline mode throughout, and refer the reader to Jouvet et al. (2026) and Rosier and Jouvet (2026) for details.

Equation (1) makes the central challenge explicit: the geometry evolution depends jointly on the unknown smb and on the ice dynamics through the flux $\mathbf{q}$. Inferring the smb from an observed geometry change therefore requires the dynamical state to be fixed beforehand; otherwise an error in the dynamics would be aliased into the inferred smb. An inconsistent initial state triggers an initialisation shock (Seroussi et al., 2011) that our short (up to ten-year) windows cannot absorb, unlike long spin-up

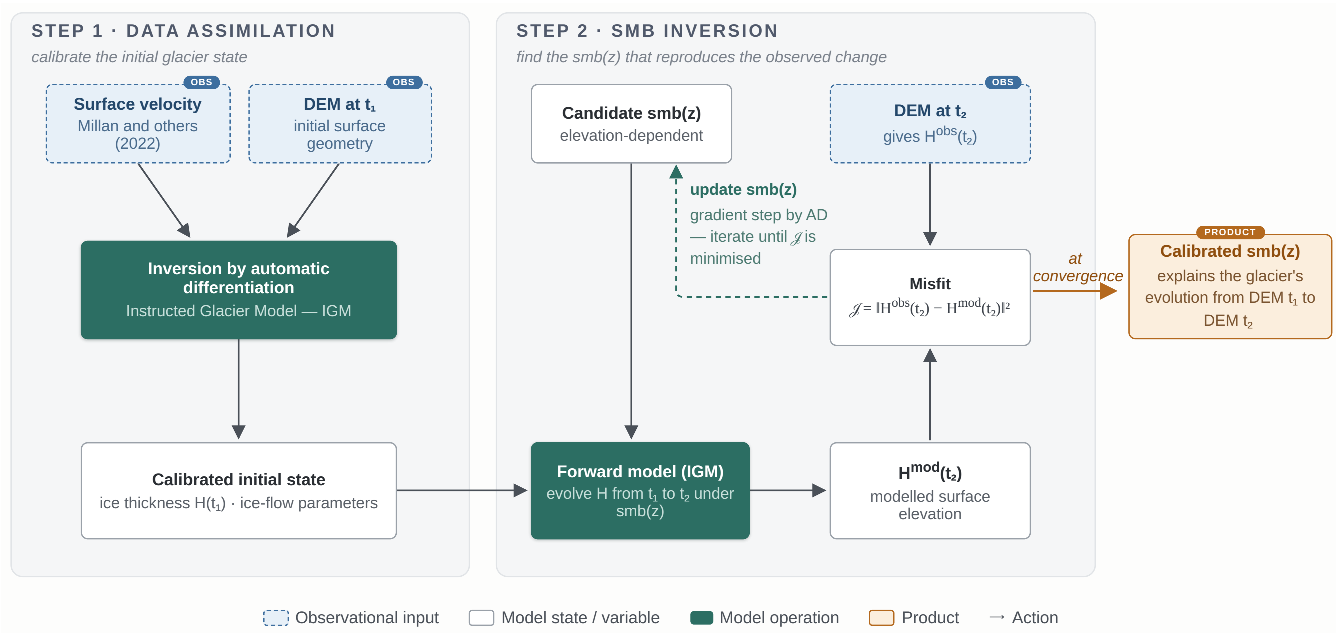

Figure 1. Overview of the two-step SMB-inference workflow. *Step 1 (data assimilation):* the observed surface velocity and the start-epoch DEM are assimilated to recover the ice thickness H_{t_1}—and hence the bedrock—while the scalar flow parameters (τ_{ref}, A) are selected by an ensemble search that minimises the surface-velocity misfit. *Step 2 (SMB inference):* with the calibrated state held fixed, the transient model is integrated from the start DEM to the end DEM under a candidate elevation-dependent profile $smb(z)$; reverse-mode automatic differentiation through the whole forward run yields the profile whose modelled end-of-window geometry best reproduces the observed one. The ice-flow emulator (frozen weights) supplies the depth-averaged velocity at every step.

simulations. Our pipeline reflects this dependence directly (Fig. 1): (i) we first calibrate the dynamical state by data assimilation against a surface-velocity snapshot near t_1, inverting the ice-thickness field H (Sect. 2.3). (ii) Holding the calibrated state fixed, we then infer the smb: starting from the t_1 DEM, we integrate (1) forward to t_2 and seek the smb whose simulated geometry best matches the observed one at t_2 (Sect. 2.4).

2.2 Data

We evaluate the method on three Alpine glaciers that span a wide range of size and are among the most intensively surveyed in the world (Table 1): Great Aletsch (Swiss Alps), the largest glacier of the Alps ($\approx$80 km^2); Rhône (Swiss Alps), a mid-sized valley glacier ($\approx$15 km^2) that feeds the eponymous river; and Argentière (French Alps, Mont-Blanc massif), a smaller, steep and fast-flowing valley glacier ($\approx$12 km^2). Each combines repeat DEMs, satellite surface-velocity fields and a decades-long in situ stake network—maintained by swisstopo and GLAMOS in Switzerland and by GLACIOCLIM in France—which supply both the fields that drive the inversion and the ground truth that validates it.

Table 1. Input and validation data for the three study glaciers. DEM epochs bracket the inversion window and velocities drive the data assimilation. The stake column gives the number of continuous-record locations averaged for the sRMSE.

	Argentière	Great Aletsch	Rhône
Window	2012–2021	2009–2017	2007–2016
Surface DEMs	Pléiades	swisstopo	swisstopo
Velocity		(Millan et al., 2022)	
Stake locations	25	2	5

For each glacier the pipeline requires two surface DEMs bracketing the inversion window—whose difference is the geometry change to be explained—and a surface-velocity field for the data assimilation (Sect. 2.3). Ice-thickness soundings are used only where available, as an optional refinement (Appendix A1). For Argentière the two DEMs are the 2012 and 2021 Pléiades models Kneib et al. (2024); for Great Aletsch (2009, 2017) and Rhône (2007, 2016) they are swisstopo geodetic DEMs. We resample all fields onto a common $100\,\mathrm{m}$ grid. Surface velocities are the satellite feature-tracking mosaics of Millan et al. (2022), built from Sentinel-2 and Landsat-8 image pairs over 2017–2018. Because alpine surface velocities vary slowly relative to the decadal DEM windows (Dehecq et al., 2019), this single field serves as a representative snapshot for the thickness calibration.

For validation we use in situ stake balances, kept strictly independent of the inversion. Rather than compare against every raw stake reading—noisy and irregularly sampled in space and time—we retain only locations with a continuous multi-year record, average each into a single balance value (± interannual standard deviation), and evaluate the inferred smb against these averages. This yields two such locations on Aletsch and five on Rhône; on Argentière the denser GLACIOCLIM network provides the published 2012–2021 mean balance at each of 25 stakes.

2.3 Ice-dynamics calibration via data assimilation

The first step of the pipeline uses two of the observational fields introduced above—the surface velocity and the t_1 DEM—to recover the ice-thickness field that places the glacier in a dynamically consistent state, free of the initialisation shock described in Sect. 2.1.

We use IGM's existing data-assimilation module (Jouvet, 2023) to invert for the thickness H by minimising the regularised misfit

$$\mathcal{J}_{\mathrm{DA}}(H) = \alpha_u \left\| \mathbf{u}_s(H) - \mathbf{u}_s^{\mathrm{obs}} \right\|^2 + \mathcal{R}(H), \tag{2}$$

where $\mathbf{u}_s(H)$ is the modelled surface velocity returned by the emulator (Rosier and Jouvet, 2026), $\mathbf{u}_s^{\mathrm{obs}}$ is the observed surface velocity (Millan et al., 2022), and $\mathcal{R}$ is a Tikhonov smoothness regulariser on H. The smoothness weight is set at the corner of the resulting L-curve (Hansen, 1992). The minimisation is solved by L-BFGS (Liu and Nocedal, 1989); we refer the reader to Jouvet (2023) for the full formulation, weighting strategy and convergence behaviour.

The emulated flow regime is not set by the geometry alone: it also depends on two scalar physical parameters, the basal-motion parameter τ_{ref} and the Arrhenius factor A. Here τ_{ref} is the reference basal shear stress in Weertman's sliding law (Weertman, 1957)—the shear stress sustaining a reference basal speed of $100\,\mathrm{m\,yr^{-1}}$, in MPa—so that a smaller τ_{ref} corresponds to easier sliding. Rather than invert these pointwise, we fix them beforehand by an ensemble search: candidate (τ_{ref}, A) pairs are drawn log-uniformly from $[0.05, 2.0]\,\mathrm{MPa}$ and $[30, 200]\,\mathrm{MPa^{-3}\,yr^{-1}}$ by a Tree-structured Parzen Estimator (TPE; Bergstra et al., 2011), as implemented in Optuna (Akiba et al., 2019), and each pair is passed through the data assimilation; we retain the pair that minimises the surface-velocity misfit, so the assimilation itself inverts the thickness H alone. We apply this identical procedure to all three glaciers, which keeps the pipeline free of glacier-specific tuning.

Data assimilation inverts the thickness at the start epoch t_1, which we denote H_{t_1}. Finding the thickness fixes the bedrock,

$$z_b(\mathbf{x}) = z_s^{\mathrm{obs}}(\mathbf{x}, t_1) - H_{t_1}(\mathbf{x}), \tag{3}$$

which provides the substrate on which the transient run subsequently evolves. Only once the calibrated state matches the observed velocity to within observational uncertainty do we accept it and proceed to the SMB inversion.

2.4 SMB inference by automatic differentiation

2.4.1 Observation target.

With the dynamical state calibrated (Sect. 2.3), the smb inversion is driven by the geometry change between the two DEM epochs. The bedrock is already fixed through Eq. (3); combining it with the observed end-of-window surface DEM gives the target thickness

$$H^{\mathrm{obs}}(\mathbf{x}, t_2) = z_s^{\mathrm{obs}}(\mathbf{x}, t_2) - z_b(\mathbf{x}). \tag{4}$$

The two surface DEMs $z_s^{\mathrm{obs}}(\cdot, t_1)$ and $z_s^{\mathrm{obs}}(\cdot, t_2)$ are the glacier-specific two-epoch elevation models of Sect. 2.2; their difference defines the geometry change the inversion must reproduce.

2.4.2 Forward model.

With the calibrated thickness H_{t_1} and the flow scalars (τ_{ref}, A) held fixed, the glacier is evolved from t_1 to t_2 by integrating (1) with an explicit forward-Euler scheme,

$$H^{n+1} = \max\big(0,\ H^n + \Delta t^n [\, smb^n - \nabla \cdot \mathbf{q}^n \,]\big), \tag{5}$$

where the non-negativity projection prevents spurious negative thicknesses near the margin. The integration starts from $H^0 = H_{t_1}$ and advances until the accumulated time reaches t_2; the number of steps N_t is therefore not prescribed but follows from the step sizes selected along the way. Each step is limited by a Courant–Friedrichs–Lewy condition,

$$\Delta t^n = \min\Big(c_{\mathrm{cfl}}\, \Delta x \big/ \max_{\Omega} \|\bar{\mathbf{u}}^n\|,\ \ \Delta t_{\max}\Big), \tag{6}$$

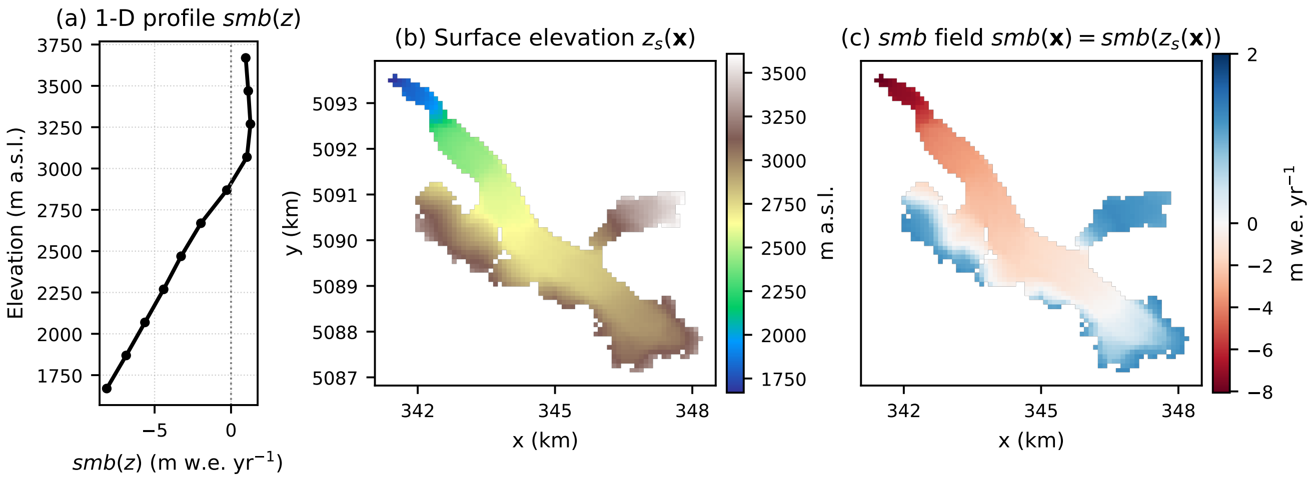

Figure 2. Mapping the one-dimensional profile to the two-dimensional source term, $smb(\mathbf{x}) = smb(z_s(\mathbf{x}))$, on Argentière. (a) The inferred elevation profile $smb(z)$; (b) the surface elevation $z_s(\mathbf{x})$; and (c) the resulting 2D smb field, obtained by evaluating the profile at the surface elevation of every grid cell.

with $c_{\text{cfl}} = 0.2 < 1$ ensuring stability and $\Delta t_{\max} = 1\,\text{yr}$ capping the step where the ice is slow. The flux divergence is evaluated on the staggered grid with a slope-limited scheme, which suppresses spurious oscillations at the steep thickness gradients near the margin. At every step the surface is updated as $z_s^n = z_b + H^n$ and $\bar{\mathbf{u}}$ is recomputed by the emulator from the current (H, z_s), so the flux divergence evolves with the geometry rather than being held at its initial-snapshot value. Conventional flux-divergence methods (Miles et al., 2021; Kneib et al., 2024), which we retain as a reference in Sect. 3, instead evaluate $smb = \partial H/\partial t + \nabla \cdot \mathbf{q}$ once from the calibrated state; because the divergence of a flux built from two independently measured fields is itself dominated by small-scale noise, such estimates require *post hoc* spatial smoothing.

2.4.3 SMB parametrisation.

Rather than treat the smb as an independent unknown at every grid cell, which would leave the inversion severely underdetermined, we parametrise it as a one-dimensional profile $smb(z)$, a function of surface elevation only. This reflects the first-order elevation control on alpine smb and collapses the unknown from $\mathcal{O}(10^4)$ pixels to $N = \mathcal{O}(10)$ elevation bins, which regularises the otherwise severely ill-posed inversion without site-specific smoothing length scales or density assumptions. The profile is discretised on uniformly spaced elevation bins, $200\,\text{m}$ wide for all three glaciers, by the values $\mathbf{m} = (m_1, \ldots, m_N)$, and it is mapped to the two-dimensional source term in (1) by evaluating it at the current surface elevation, $smb(\mathbf{x}, t) = smb(z_s(\mathbf{x}, t))$, with piecewise-linear interpolation between bins (Fig. 2). The parametrisation is deliberately minimal: our aim is an smb that explains the observed geometry change, not the resolution of its fine spatial structure.

2.4.4 Inverse problem.

We seek the smb profile $\mathbf{m}^\star$ that minimises the misfit between the simulated end-of-window thickness $H^{\mathrm{sim}}(\cdot, t_2; \mathbf{m})$ and the target $H^{\mathrm{obs}}(\cdot, t_2)$ from (4):

$$\mathcal{J}(\mathbf{m}) = \sqrt{\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} \left(H^{\mathrm{sim}} - H^{\mathrm{obs}}\right)^2} + \mathcal{R}(\mathbf{m}), \tag{7}$$

where the first term is the RMS geometry misfit and $\mathcal{R}(\mathbf{m}) = \lambda \sum_i (m_{i-1} - 2m_i + m_{i+1})^2$ is a Tikhonov smoothness penalty on the squared discrete curvature of the profile. The weight λ carries a direct physical reading: it sets how much the *smb* may change from one elevation bin to the next, and hence how strongly the recovered gradient is smoothed. We use $\lambda = 0.1$ throughout; the same value proved optimal for all three glaciers, which suggests it may transfer to alpine glaciers more generally. The minimisation uses L-BFGS (Liu and Nocedal, 1989) with $M = 10$ correction pairs and a Hager–Zhang line search; the low dimensionality of $\mathbf{m}$ makes the implicit Hessian approximation accurate, and convergence is reached in $\mathcal{O}(10^2)$ function evaluations.

The gradient $\nabla_{\mathbf{m}} \mathcal{J}$ is obtained by reverse-mode automatic differentiation (Baydin et al., 2018) through the entire forward integration (1)–(5): every operation in the forward map—the emulator evaluation, the slope-limited divergence, the explicit Euler update and the piecewise-linear elevation interpolation—is a differentiable TensorFlow (Abadi et al., 2016) operation, so the chain rule applies uniformly across the forward graph and the sensitivity of the end-of-window geometry to each bin value is propagated exactly through every time step.

The principal cost is memory. Automatic differentiation must retain intermediate quantities from the forward run in order to traverse it backwards, and a decade-long integration exceeds the memory of a single GPU. We therefore use gradient checkpointing (Griewank and Walther, 2000; Chen et al., 2016): instead of storing the internal quantities of every step, we keep only the glacier state (H, z_s) at the start of each step and recompute that step's internals when the backward pass reaches it. The saving is substantial because the emulator evaluation—by far the heaviest part of a step—is never stored, only replayed; the price is one additional evaluation of each step per gradient.

3 Results

We evaluate the method on three Alpine glaciers of contrasting size—Argentière, Great Aletsch and Rhône—each processed through the same glacier-agnostic two-step pipeline: the data assimilation inverts the ice thickness while the basal-motion parameter τ_{ref} is fixed by an ensemble search to the scalar that minimises the surface-velocity misfit, after which the *smb* is inferred. No per-glacier retuning of the regularisation, optimiser or parametrisation is applied, and none of the three uses a thickness survey. Argentière additionally carries the state-of-the-art distributed reconstruction of Kneib et al. (2024), against which we compare our velocity-only result directly.

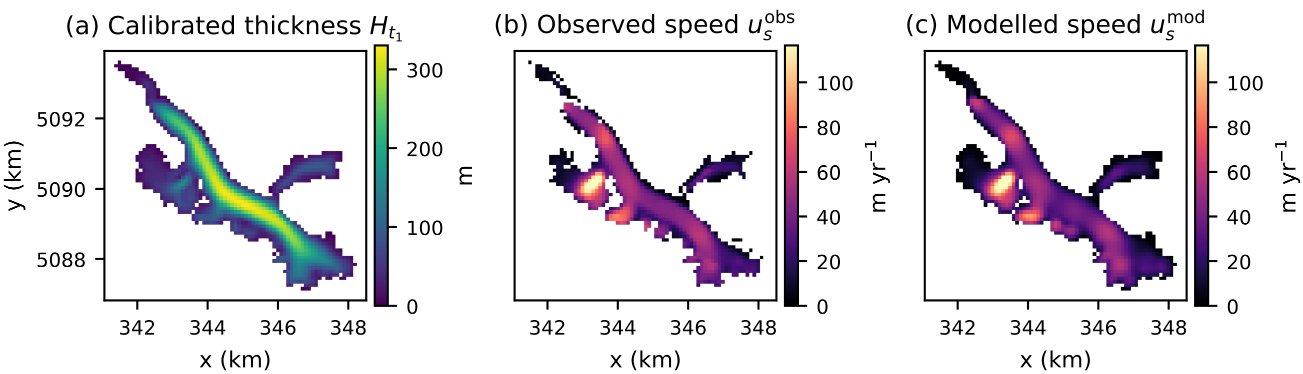

Figure 3. Calibrated data-assimilation state on Argentière at the start epoch, for the adopted velocity-only run ($\tau_{\mathrm{ref}} = 0.27\,\mathrm{MPa}$). (a) Calibrated ice thickness H_{t_1}; (b) observed surface speed u_s^{obs} (Millan et al., 2022); (c) modelled surface speed u_s^{mod}, in close agreement with the observations.

3.1 Calibrated ice-dynamics states

For every glacier the data assimilation converges within $\mathcal{O}(10^2)$ L-BFGS iterations to a thickness field whose modelled surface velocity matches the observations to a root-mean-square residual of a few $\mathrm{m\,yr^{-1}}$—2.4 on Argentière, 11 on Great Aletsch and 6.8 on Rhône at the adopted τ_{ref}—one to two orders of magnitude below typical trunk velocities (locally exceeding 100–$150\,\mathrm{m\,yr^{-1}}$). The calibrated state fixes the bedrock through Eq. (3) and provides the starting geometry H_{t_1} for the transient forward run (Sect. 2.4).

Figure 3 shows the calibrated state on Argentière: the recovered ice-thickness field (a), and the close match between the observed (b) and modelled (c) surface speed, whose differences are small and spatially unstructured.

As a final check, we confirm the calibrated state is free of the initialisation shock of Sect. 2.1: integrated forward for one year under zero smb, it produces no spurious thickness drift, so the glacier is in mechanical equilibrium and any subsequent geometry change is attributable to the prescribed smb rather than to residual model imbalance.

3.2 Inferred SMB and validation

Holding the calibrated state $(H_{t_1}, \tau_{\mathrm{ref}}, A)$ fixed, we run the transient forward simulation over each glacier's window and invert for the elevation-resolved smb profile $smb(z)$ that minimises the geometry-misfit cost (Eq. 7). On all three glaciers the retrieved profile increases monotonically across the full altitudinal range, from strongly negative values at the tongue to a positive accumulation regime in the upper basin.

On Argentière (2012–2021) the profile reproduces the observed altitudinal pattern of the GLACIOCLIM stakes with a root-mean-square error of $0.65\,\mathrm{m\,w.e.\,yr^{-1}}$ (Fig. 4a, Table 2). This is obtained with the generic velocity-only pipeline—no thickness survey, no per-glacier tuning, no *post hoc* smoothing—yet it is on par with the best physically-based reconstructions of Kneib

Table 2. Validation of the inferred SMB against the GLACIOCLIM stake network on Argentière Glacier (main trunk + Améthystes tributary) over 2012–2021. The three baselines are the modelling approaches of Kneib et al. (2024), evaluated against the same stakes; all three use ice-thickness data, whereas our result uses surface velocity only. The GPR-assisted field inversion, which improves further on these, is reported in Appendix A1.

Method	RMSE [$\mathrm{m\,w.e.\,yr^{-1}}$]
F2019 (Kneib et al., 2024)	0.50
SIA (Kneib et al., 2024)	0.67
IGM-distributed (Kneib et al., 2024)	0.85
Ours (velocity only, scalar τ_{ref})	**0.65**

et al. (2024) on the same stakes, whose SIA and IGM-distributed approaches give 0.67 and $0.85\,\mathrm{m\,w.e.\,yr^{-1}}$ and whose flux-gate F2019 estimate gives 0.50, all of which additionally require ice-thickness data. Argentière is also covered by accurate GPR thickness soundings; repeating the experiment with them—inverting τ_{ref} as a spatial field—lowers the error further and surpasses every published reconstruction on this site, as detailed in Appendix A1. Because the inversion variable is one-dimensional, horizontal heterogeneity within an elevation band—notably the avalanche-fed accumulation hotspots reported by Kneib et al. (2024)—is not resolved; this deliberate trade-off, which buys a tuning-free and transferable inversion, is taken up in Sect. 4.

The same pipeline, applied without modification to Great Aletsch (2009–2017) and Rhône (2007–2016), yields equally smooth, monotonic profiles that reproduce the independent stake averages—two continuous-record locations on Aletsch and five on Rhône (Sect. 2.2)—with root-mean-square errors of 0.61 and $0.32\,\mathrm{m\,w.e.\,yr^{-1}}$ (Fig. 4b,c). These are on par with the Argentière result despite the absence of any thickness constraint, which we read less as a coincidence than as evidence that the velocity-calibrated dynamical state carries the elevation structure of the SMB; whether the robustness holds across a larger and more varied sample than three large glaciers is an open question we return to in Sect. 4.

Figure 4 makes concrete the mass-conservation argument of Sect. 1.1. The dashed curve is the SMB the continuity equation yields when the flux divergence is taken as stationary, exactly as in flux-divergence inversions. On every glacier this estimate is markedly noisier than the inferred profile and diverges in the ablation zone, where the stationary divergence is least representative, raising the stake RMSE to 2.14 (Argentière), 1.59 (Aletsch) and $1.63\,\mathrm{m\,w.e.\,yr^{-1}}$ (Rhône)—a factor of 2.6 to 5.1 worse than our transient inversion. Because our forward model integrates the mass-conservation law directly and recomputes the flux divergence from the evolving geometry at every step, the inferred profile requires no spatial smoothing and remains mass-consistent throughout.

3.3 Sensitivity to the basal-motion parameter

The basal-motion parameter τ_{ref} is the dynamical quantity least constrained on a glacier without thickness or stake data, where its value rests on the surface-velocity misfit (*velsurf*) alone. It is also intrinsically hard to separate from the ice thickness: a more slippery bed (smaller τ_{ref}) inferred together with thinner ice reproduces almost the same surface velocity—and hence the

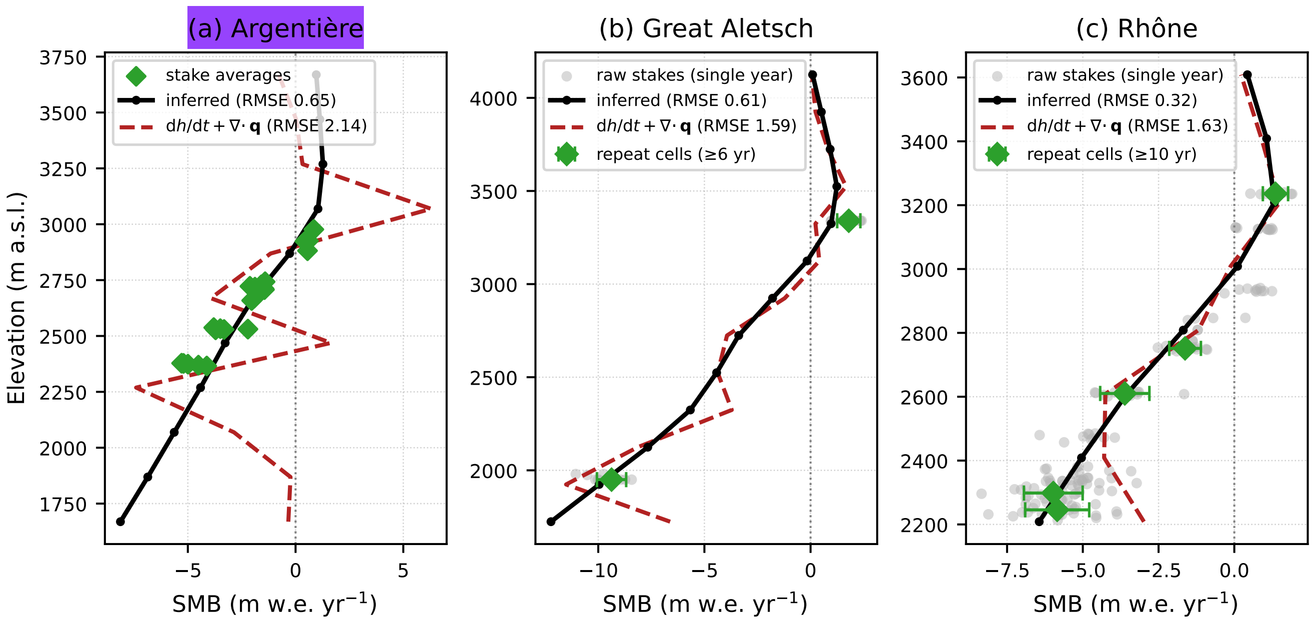

Figure 4. Inferred SMB profile (black) for (a) Argentière, (b) Great Aletsch and (c) Rhône, compared against in situ stake measurements and against the continuity estimate $\mathrm{d}h/\mathrm{d}t + \nabla\cdot\mathbf{q}$ (dashed) obtained with the *stationary* data-assimilation flux divergence. Green diamonds are the continuous-record stake averages ($\pm$ interannual standard deviation), the validation targets (faint grey dots on Aletsch and Rhône are the raw single-year readings). The inferred profile tracks the averages across the whole elevation range ($\mathrm{RMSE} = 0.65$, 0.61 and $0.32\,\mathrm{m\,w.e.\,yr^{-1}}$), whereas the stationary-flux estimate departs strongly in the ablation zone (2.14, 1.59 and 1.63).

same ice flux—as a stiffer bed carrying thicker ice, so many (τ_{ref}, H) pairs fit the observations nearly equally well. For the smb inversion this ambiguity is not fatal, because what drives the geometry change is not the pair itself but the flux divergence $\nabla\cdot\mathbf{q}$ it produces. To see how the residual freedom in τ_{ref} propagates into the inferred smb, we repeated the smb inversion across a wide range of τ_{ref} on all three glaciers (Fig. 5), using the independent stake RMSE as ground truth against *velsurf*, the only selector available operationally.

Two features stand out. First, *velsurf* reliably rejects the strongly biased runs: the profiles that depart most from the stakes are also those with the worst velocity misfit (Fig. 5). Pooled across the three glaciers, higher velocity misfit goes with higher stake error (Fig. 6; Pearson $r = 0.69$, $p < 10^{-2}$). Second, once these high-misfit runs are removed, the surviving low-*velsurf* runs no longer map one-to-one onto stake error: because a wide range of (τ_{ref}, H) pairs share nearly the same flux, velocity cannot single out one value of τ_{ref}, and on Rhône the stake RMSE even varies non-monotonically among them (0.32–$1.25\,\mathrm{m\,w.e.\,yr^{-1}}$). Crucially, this residual freedom is confined to the ablation zone: across the low-misfit runs the accumulation-area profile is essentially unchanged, and only the tongue smb—and with it the equilibrium-line altitude—shifts from run to run (Fig. 5).

The practical consequence is that the method does not deliver a single "best" profile but a family of low-*velsurf* solutions that converge to the same accumulation-area gradient and fan out only at the tongue (Fig. 5, Fig. 6). For a stake-free glacier

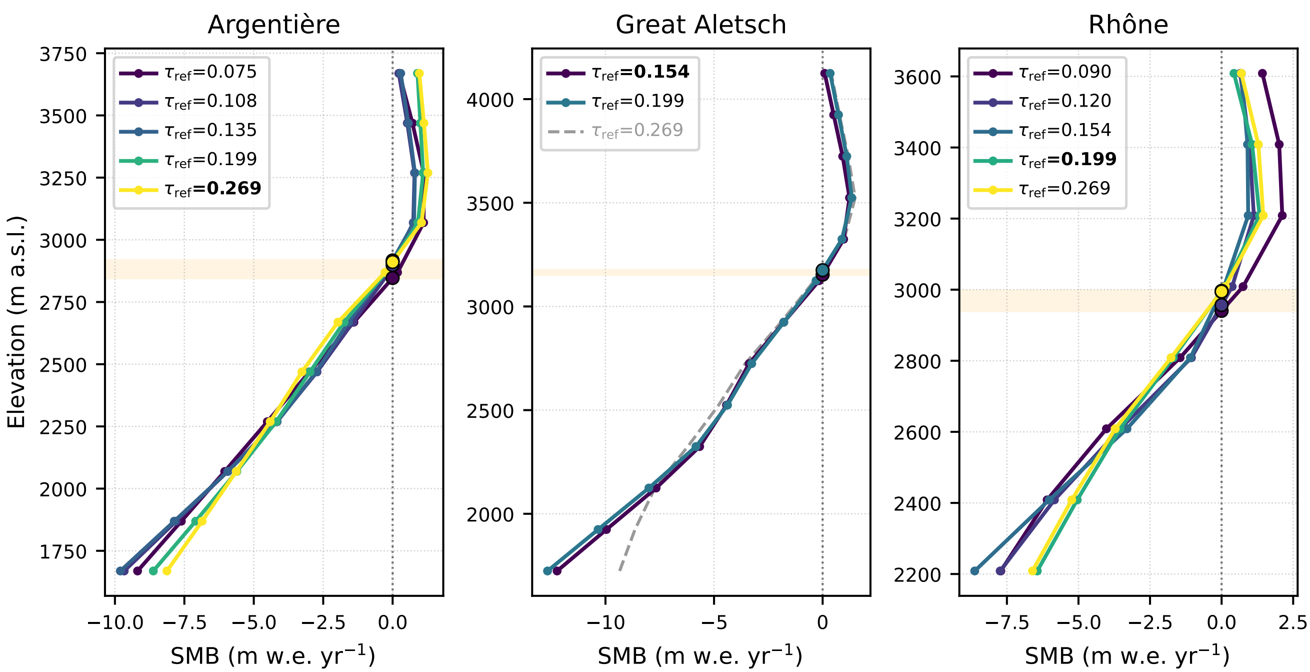

Figure 5. Inferred SMB profiles across a range of the basal-motion parameter τ_{ref} for (a) Argentière, (b) Great Aletsch and (c) Rhône; colour encodes τ_{ref} and the adopted profile is drawn bold. Within the low-velocity-misfit basin the profiles coincide in the accumulation area and diverge only at the ablation tongue, so the residual uncertainty in τ_{ref} translates into an equilibrium-line-altitude band (shaded) rather than a whole-profile bias; the runs the velocity misfit rejects (dashed) depart most strongly.

this family is the result: the low-misfit runs together define an smb and equilibrium-line uncertainty band—narrow in the accumulation area, widening towards the tongue. How this band behaves at regional scale, and how it might be tightened, we take up in Sect. 4.

4 Discussion

Separating surface mass balance from dynamic thinning in an observed surface-elevation change is a notoriously ill-posed problem, yet it is exactly what deriving [illegible] from remote sensing demands. We have addressed it with a transient, differentiable inversion: a state-of-the-art assimilation of surface ice speed fixes the dynamical state, and a novel automatic-differentiation scheme then inverts the multi-temporal geometry change for the elevation-resolved SMB. On three Alpine glaciers of contrasting size the method reproduces the in situ stake balances to root-mean-square errors of 0.65 (Argentière), 0.61 (Great Aletsch) and $0.32\,\mathrm{m\,w.e.\,yr^{-1}}$ (Rhône), on par with the best physically-based reconstructions of Kneib et al. (2024) on the same Argentière stakes and, where GPR thickness is assimilated, surpassing them (Appendix A1). Two features set the approach apart from that state of the art. First, the flux divergence is recomputed from the evolving geometry at every time step rather than

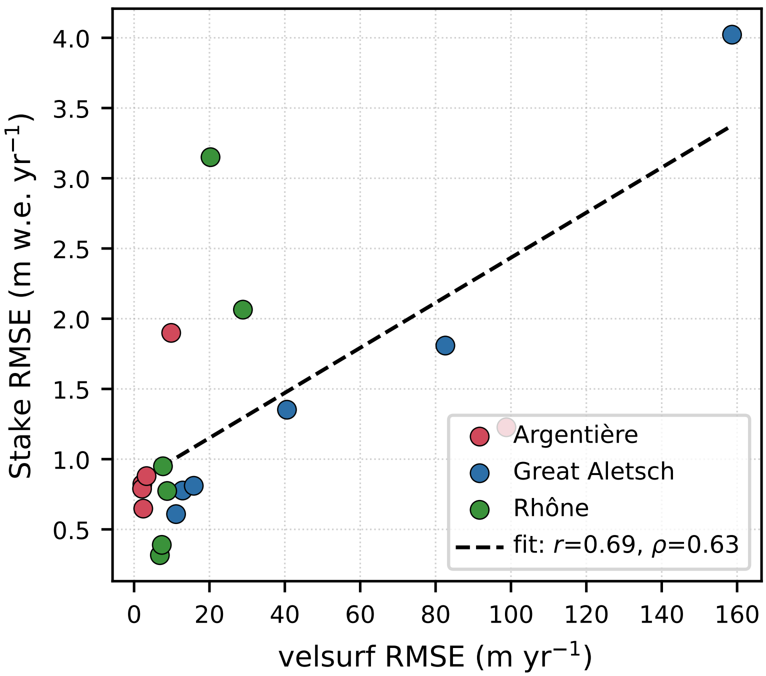

Figure 6. Data-assimilation surface-velocity misfit (*velsurf* RMSE) against the independent stake RMSE, pooled over the τ_{ref} sweep (six runs per glacier, all three glaciers). Higher velocity misfit is associated with higher stake error (Pearson $r = 0.69$, Spearman $\rho = 0.63$), so the velocity misfit—the only selector available without stakes—carries usable information about the accuracy of the inferred SMB.

held at a single stationary snapshot, so the inferred SMB is dynamically consistent and avoids the ablation-zone bias that makes stationary-flux estimates a factor of two to five worse against the same stakes. Second, the inversion needs only surface velocity—no ice-thickness survey and no per-glacier tuning—and proved largely insensitive to the rate factor A, so a single glacier-independent pipeline transfers across all three sites.

The clearest methodological gain is in how mass conservation is enforced. Flux-divergence inversions form the SMB by differentiating the observed thickness–velocity flux and combining it with the elevation-change field; because that divergence is the spatial derivative of a product of two noisy fields, it must be smoothed before use, and the smoothing—as Kneib et al. (2024) document—breaks the very mass-conservation constraint the method rests on, leaving a spurious glacier-wide flux-divergence residual of order 0.1–$0.7\,\mathrm{m\,yr^{-1}}$ that they close by redistributing the excess or deficit uniformly across the glacier. Such a homogeneous redistribution is a pragmatic fix rather than a physical statement: it spreads a computational residual over cells that did not produce it. Our formulation removes the need for either step. Because we integrate the mass-conservation equation forward and evaluate the flux divergence on the modelled geometry with a conservative, slope-limited scheme, mass is conserved by construction at every step; the inferred SMB field requires no post-hoc smoothing and no ad-hoc redistribution, and the divergence it implies is guaranteed consistent with the simulated geometry change.

The method's principal limitation is its one-dimensional parametrisation: horizontal variability within an elevation band—such as the avalanche-fed accumulation hotspots resolved by Kneib et al. (2024)—is deliberately not recovered. This buys a transferable, tuning-free inversion and is not intrinsic to the scheme, which admits richer descriptions (an aspect or debris-cover dependence, say) wherever the data warrant them. A second limitation is the thickness–sliding ambiguity of Sect. 3.3: surface velocity constrains the basal-motion parameter only to a band, and the resulting freedom, negligible in the accumulation area, opens into an equilibrium-line spread at the tongue. Two further dependencies are shared with flux-based methods. Accuracy is ultimately set by the input velocity and DEMs: Kneib et al. (2024) find that substituting the coarser global velocity mosaic of Millan et al. (2022) into their reconstruction degrades the stake agreement to $\sim$1.2–1.3 $\mathrm{m\,w.e.\,yr^{-1}}$, so the velocity field can itself become limiting—that our transient inversion attains 0.32–0.65 $\mathrm{m\,w.e.\,yr^{-1}}$ using precisely that global product is encouraging, but its robustness to velocity error should not be overstated. Finally, like all geodetic estimates the method converts a volume change to a mass balance under an assumed ice density and neglects evolving firn compaction; where the firn column is thick or densifying, this introduces a systematic offset.

Taken together, these properties point towards regional-scale application. Because the pipeline requires no thickness survey, no stake calibration and no site-specific tuning, it can in principle be applied wherever multi-temporal DEMs and a surface-velocity field exist—and both are approaching near-global coverage, through the elevation-change archive of Hugonnet et al. (2021) and expanding velocity products (Millan et al., 2022). Its accuracy will track that of these inputs, so the highest fidelity will come from dedicated high-resolution DEMs; but the tuning-free character makes a continuous, distributed SMB inversion over many glaciers a realistic next step. Whether the robustness shown here on three large, well-monitored glaciers persists across the smaller and more varied population that dominates regional water resources is the key open question, and the natural direction for future work.

5 Conclusions

We have presented a transient, differentiable inversion that recovers the distributed surface mass balance of a glacier from two surface DEMs and a single surface-velocity field, without ice-thickness measurements or per-glacier tuning. Surface velocity is assimilated to fix a dynamically consistent state, and the mass-conservation equation is then integrated forward with automatic differentiation to infer the elevation-resolved SMB whose modelled geometry change best matches observation. Applied through one glacier-independent pipeline to three Alpine glaciers of contrasting size, the method reproduces in situ stake balances to 0.32–0.65 $\mathrm{m\,w.e.\,yr^{-1}}$—on par with, and where GPR thickness is available surpassing, the state-of-the-art reconstruction of Kneib et al. (2024), and a factor of two to five better than the stationary-flux estimates it replaces.

The central advance is that mass conservation is enforced by construction: because the flux divergence is recomputed from the evolving geometry at every step, the inferred SMB needs neither the post-hoc smoothing nor the ad-hoc mass redistribution that flux-divergence methods require, and stays dynamically consistent over the whole observation window. Its accuracy is limited chiefly by the input velocity and DEMs and by the one-dimensional elevation parametrisation, which trades sub-band spatial detail for transferability. Because the method needs no calibration data beyond fields that are approaching near-global coverage, it opens a realistic path towards continuous, regional-scale SMB inversion; establishing its robustness across the smaller and more varied glacier population that dominates regional water resources is the natural next step.

Appendix A: Appendix

A1 GPR-assisted basal-motion-field inversion on Argentière

The main analysis fixes the basal-motion parameter τ_{ref} to a single scalar, so that all three glaciers are treated identically and no thickness survey is required. Where dense auxiliary data *do* exist, however, the same differentiable framework can exploit them to sharpen the dynamical state. Argentière is covered by repeated ground-penetrating-radar (GPR) soundings of ice thickness (Rabatel et al., 2018); here we show that assimilating them, and relaxing τ_{ref} from a scalar into a spatial field $\tau_{\mathrm{ref}}(\mathbf{x})$ inverted jointly with the thickness, further improves the fit and surpasses the state of the art.

The GPR soundings supply an independent thickness constraint (the second term of Eq. 2, over the survey footprint Ω_{thk}) that makes the joint inversion identifiable. The data assimilation then retrieves both the ice-thickness field H_{t_1} and the basal-motion field $\tau_{\mathrm{ref}}(\mathbf{x})$ that together reproduce the observed surface velocity and the GPR transects, converging within 10^2 L-BFGS iterations (Fig. A1). Modelled velocities match the observations to a root-mean-square residual of $3\,\mathrm{m\,yr^{-1}}$, and the calibrated thickness departs from the GPR transects by $\mathrm{RMS} = 12\,\mathrm{m}$ (bias $-9\,\mathrm{m}$), of the order of the GPR uncertainty itself. The recovered sliding field shows the physically expected structure—enhanced along the central flowline and reduced near the lateral margins—consistent with the finding of Kneib et al. (2024) that spatial variability in τ_{ref} helps explain the observed surface velocities.

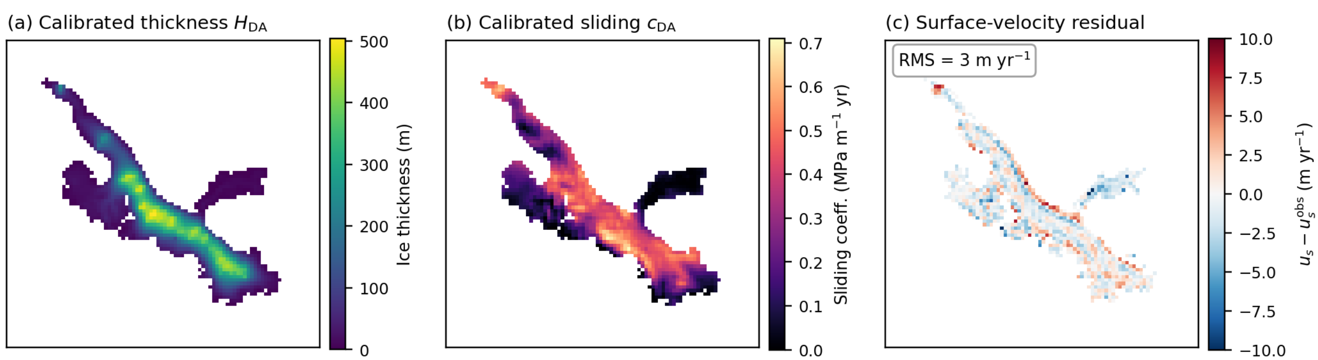

Figure A1. GPR-assisted data-assimilation state on Argentière Glacier at the start epoch. (a) Calibrated ice thickness H_{t_1} and (b) calibrated basal-motion field $\tau_{\mathrm{ref}}(\mathbf{x})$; (c) surface-velocity residual $u_s - u_s^{\mathrm{obs}}$, near zero across the glacier (overall $\mathrm{RMS} = 3\,\mathrm{m\,yr^{-1}}$; white cells lack a velocity observation).

Holding this calibrated state fixed, the transient inversion (2012–2021) yields the elevation-resolved SMB profile of Fig. A2. The simulated end-of-window thickness matches the target across the trunk to within the DEM-differencing uncertainty (Fig. A3; residual $\mathrm{RMS} = 24\,\mathrm{m}$, small and largely unbiased). Validated against the GLACIOCLIM network, the profile reproduces the observed altitudinal pattern with a root-mean-square error of $0.45\,\mathrm{m\,w.e.\,yr^{-1}}$—lower than the velocity-only result of the main text (0.65) and below every previously published distributed reconstruction on this site over this period, including

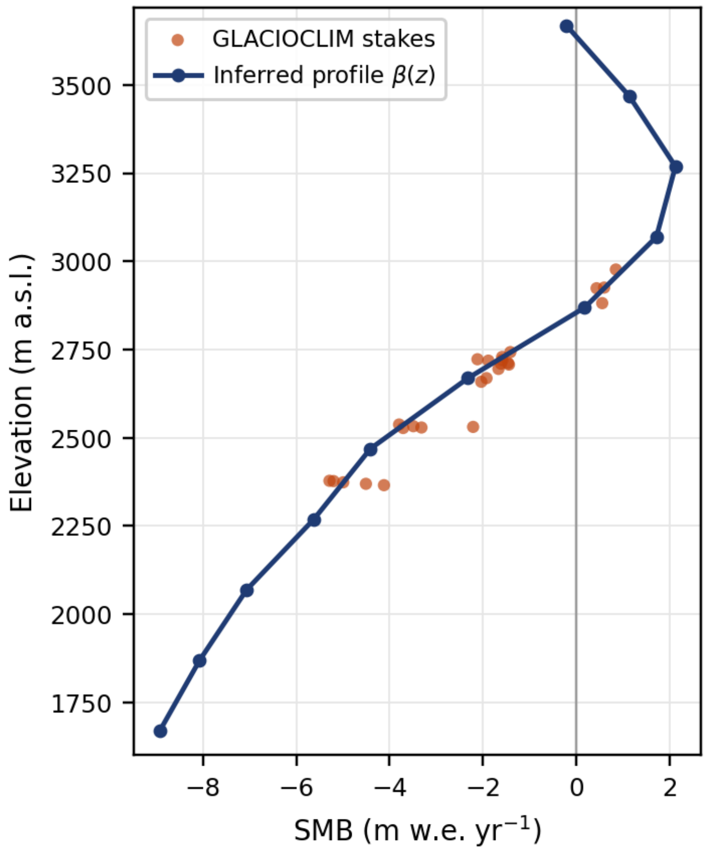

Figure A2. Inferred elevation-resolved SMB profile $\beta(z)$ (blue) for Argentière over 2012–2021 from the GPR-assisted inversion, in water equivalent. Orange dots are the 2012–2021 mean GLACIOCLIM stake balances, plotted at the elevation of each stake. The profile spans 1670–3610 m a.s.l. and matches the stakes with $\mathrm{RMSE} = 0.45\,\mathrm{m\,w.e.\,yr^{-1}}$.

all three baselines of Kneib et al. (2024) (0.50–$0.85\,\mathrm{m\,w.e.\,yr^{-1}}$; Table 2). The residual structured misfit in Fig. A3(c) reflects horizontal variability within elevation bands—notably the avalanche-fed accumulation hotspots (Kneib et al., 2024)—that the one-dimensional profile does not represent.

A2 Applicability to near-global elevation-change data

The fidelity of the inferred SMB is ultimately set by the quality of the input elevation change (Sect. 1.2). The main results use dedicated two-epoch DEMs, which deliver the highest accuracy but are not available everywhere. Near-global elevation-change products such as that of Hugonnet et al. (2021) would, by contrast, make the method applicable to essentially any glacier. To gauge the associated fidelity penalty, we repeat the inversion on Rhône—chosen for its denser stake record ($n = 5$

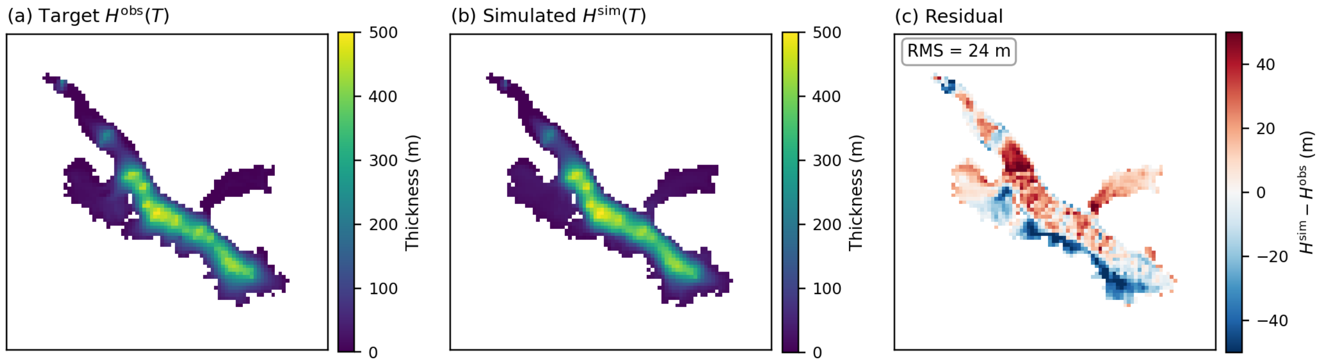

Figure A3. End-of-window ice-thickness fit on Argentière ($t_2 = 2021$), masked to the glacier footprint. (a) Target thickness $H^{\mathrm{obs}}(\cdot, t_2)$ from the observed surface DEM and the calibrated bedrock; (b) simulated thickness $H^{\mathrm{sim}}(\cdot, t_2)$ from the forward run under the inferred SMB profile; (c) their residual $H^{\mathrm{sim}} - H^{\mathrm{obs}}$ (RMS $24\,\mathrm{m}$). The residual is small and largely unbiased; its spatially structured component reflects horizontal variability within elevation bands that the one-dimensional profile does not represent.

continuous-record locations)—replacing only the geometry-change target: the observed surface-elevation change is now the Hugonnet et al. (2021) $\mathrm{d}h/\mathrm{d}t$ rate, while the velocity data assimilation and the scalar-τ_{ref} ensemble search are left unchanged.

The velocity-selected coefficient ($\tau_{\mathrm{ref}} = 0.12\,\mathrm{MPa}$, again the surface-velocity-misfit minimum) yields an inferred profile that reproduces the stake averages with a root-mean-square error of $0.70\,\mathrm{m\,w.e.\,yr^{-1}}$ (Fig. A4)—about twice the 0.32 obtained with the dedicated swisstopo DEMs (Sect. 3.2), yet still recovering the full altitudinal structure and remaining well below the stationary-flux continuity estimate ($1.79\,\mathrm{m\,w.e.\,yr^{-1}}$). The degradation is concentrated in the upper ablation and accumulation area, and reflects the stronger spatial smoothing and larger uncertainty of the near-global product relative to a dedicated DEM pair. This confirms the trade-off anticipated in Sect. 1.2: near-global elevation-change data extend the method to almost any glacier at a fidelity cost bounded by the DEM quality, so dedicated multi-temporal DEMs remain preferable where they exist. Exploring the full sliding sweep (Table A1) reinforces this: with the Hugonnet product the surface-velocity-misfit minimum coincides with the best stake fit, and the stake error is nearly flat across the whole low-misfit basin (0.70–$0.77\,\mathrm{m\,w.e.\,yr^{-1}}$), so the residual sensitivity to τ_{ref} is small next to the fidelity gap between the two elevation-change sources.

Table A1. Sensitivity of the Hugonnet-based Rhône inversion to the basal-motion parameter τ_{ref} (at $A = 78$), with the dedicated-DEM stake RMSE at matching τ_{ref} for comparison. *velsurf* is the surface-velocity misfit and ELA the equilibrium-line altitude; the stake RMSE is the validation against the $n = 5$ continuous-record averages. Column minima in bold.

τ_{ref} (MPa)	velsurf RMSE ($\mathrm{m\,yr^{-1}}$)	ELA (m a.s.l.)	Stake RMSE ($\mathrm{m\,w.e.\,yr^{-1}}$) Hugonnet	ded. DEMs
0.121	**7.56**	3136	**0.70**	0.81
0.154	8.43	3096	0.74	1.25
0.199	9.42	3084	0.77	**0.32**
0.459	15.35	3083	0.73	0.77
0.744	34.77	3183	2.66	3.15
1.078	37.43	3422	2.28	2.06

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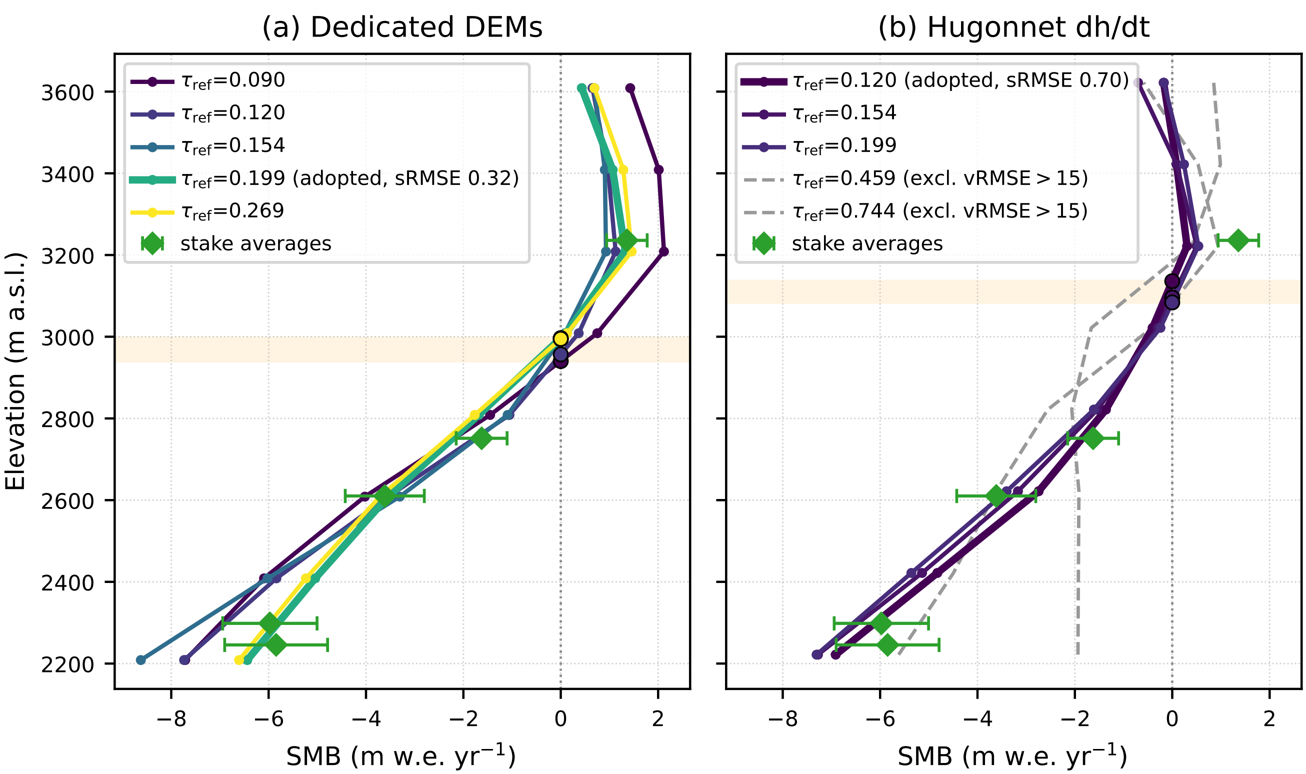

Figure A4. Sliding-coefficient sensitivity of the Rhône inversion for the two elevation-change experiments: (a) the dedicated two-epoch DEMs used in the main text and (b) the Hugonnet et al. (2021) $\mathrm{d}h/\mathrm{d}t$ product. Each panel shows the inferred SMB profile across a range of $\tau_{\rm ref}$ (colour; the adopted profile bold with its sRMSE, and runs rejected by the velocity misfit dashed), the equilibrium-line-altitude band shaded, all validated against the same continuous-record stake averages (green diamonds). Both experiments recover the elevation structure, but the Hugonnet-based fan sits higher and is biased low in the accumulation area, giving a larger stake RMSE (0.70 vs $0.32\,\mathrm{m\,w.e.\,yr^{-1}}$).

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